

NASA INTERNATIONAL SPACE APPS CHALLENGE 2026

Hunting Planet X with SPHEREx

A Public Infrared Photometry Workstation for Multi-Epoch Sky Analysis & Transient Discovery

Challenge: Planet X and SPHEREx · Team: Exulans · Track: Astrophysics & Planetary Science

Live Deployment: https://spherex-photometry.vercel.app · Open Source: github.com/Yancovert/spherex-photometry

Finding What Moves in a Billion-Object Sky

The Physical Detection Barriers:

- **Sub-Pixel Proper Motion:** At 400 AU, Planet X shifts only 0.385 arcseconds over SPHEREx's 6-month survey cadence. With a 6.20"/px detector pitch, this displacement is 0.062 pixels (< 1/16th of a single pixel).
- **Extreme Optical Dimming (1/d^4):** Reflected sunlight scales inversely with the fourth power of distance. At 400 AU, optical flux is attenuated by a factor of 2.56 x 10^10 compared to 1 AU.
- **Infrared Thermal Dominance (1/d^2):** Thermal blackbody radiation drops only as 1/d^2, peaking beyond 3.5 um (SPHEREx Bands 4-6) at equilibrium temperatures T_eq ≈ 14-40 K.
- **The Tooling Gap:** Before our work, no public, lightweight web tool existed allowing researchers and citizen scientists to blink SPHEREx epochs and extract calibrated aperture photometry.

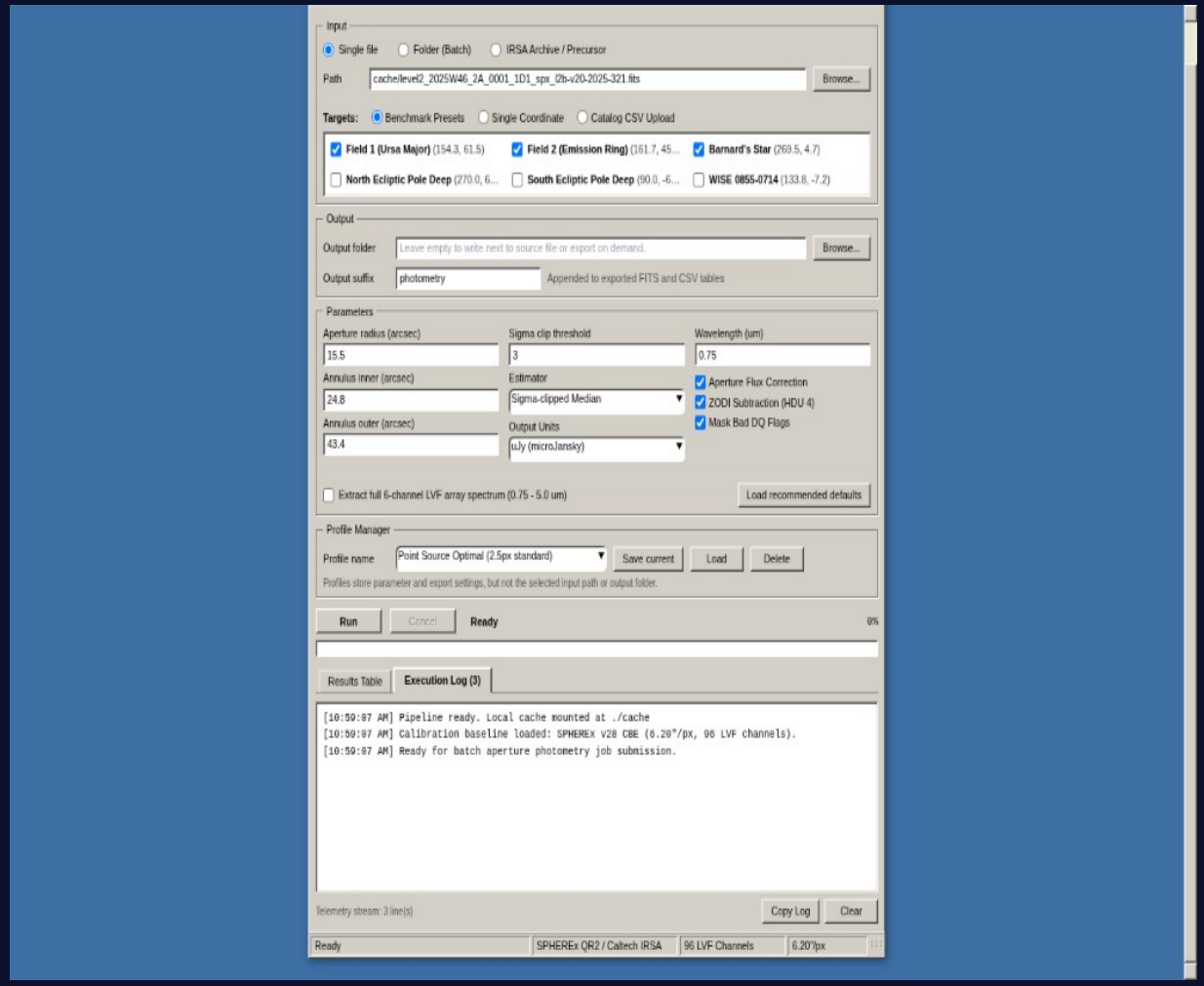


ARCHITECTURE & USER EXPERIENCE

Authentic Win32 Scientific Discovery Workstation

Zero Modern Web Bloat. Real Science:

- **Multi-Epoch Blink Comparator:** WiseView-heritage 4-epoch blinking at 1-10 Hz. Supports 8 scientific colormaps (Monochrome, Viridis, Inferno, Turbo) and asinh/linear contrast stretches.
- **Difference Imaging Mode:** Instant signed epoch differencing (E2 - E1) that cancels static stars and isolates transient and moving-body dipole signatures.
- **Batch Aperture Photometry Engine:** Sub-pixel circular aperture integration with MAD sigma-clipped sky background estimation across all 96 LVF channels.
- **Gaia DR3 & NEOSSat Cross-Matching:** Real-time coordinate queries to ESA Gaia DR3 for proper motion verification and CSA NEOSSat asteroid orbits.



MATHEMATICAL PHYSICS

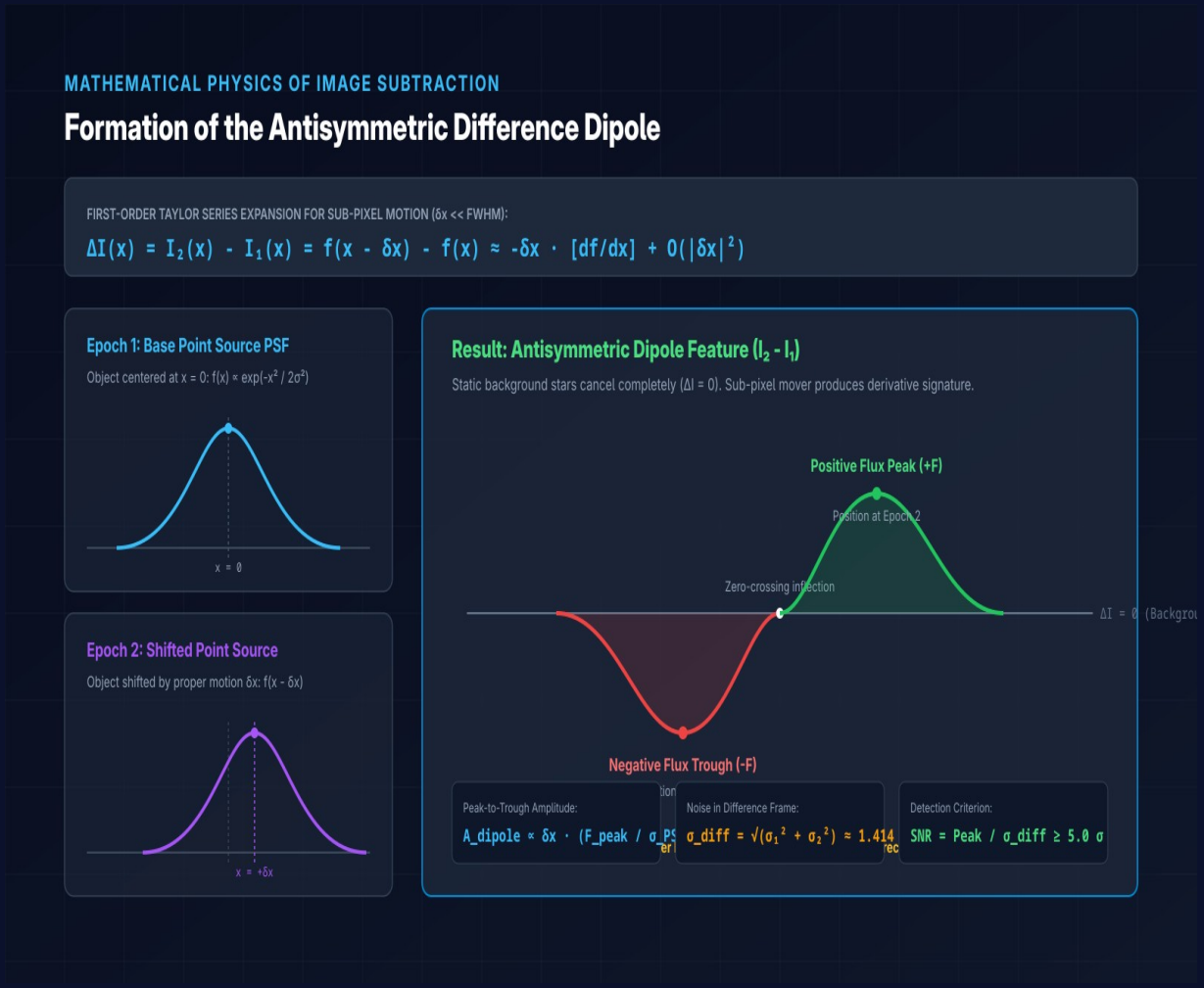
Image Differencing & Dipole Formation

First-Order Taylor Series Derivation:

- **Epoch Image Formulation:** Let I_1(x) = f(x) be the point source at Epoch 1. At Epoch 2, it shifts by proper motion dx: I_2(x) = f(x - dx).
- **Difference Image Residual:** ΔI(x) = I_2(x) - I_1(x) = f(x - dx) - f(x). For small shifts dx << FWHM, expanding via Taylor series yields:

ΔI(x) ≈ -dx · [df/dx] + O(|dx|^2)

- **The Antisymmetric Dipole:** The residual profile is proportional to the spatial derivative of the telescope PSF. This produces an antisymmetric dipole: a positive flux peak (+F) right next to a negative trough (-F).
- **Background Cancellation:** Stationary background stars have dx = 0 and vanish completely (ΔI = 0). Noise adds in quadrature: σ_diff = √(σ_1^2 + σ_2^2) ≈ 1.414 · σ_0.



PIPELINE ARCHITECTURE & CALIBRATIONS

From Raw FITS Images to Calibrated Flux Catalogs

END-TO-END COMPUTATIONAL PIPELINE

SPHEREx Scientific Photometry Architecture

1. INGESTION & WCS

SPHEREx L2 Products

- 2048 x 2048 FITS multi-extension
- Surface Brightness (MJy/sr)
- Caltech IRSA TAP API integration
- WCS coordinate projection (TAN)

`Pixel scale: 6.20"/px pitch`

2. CORE PHOTOMETRY

Sub-Pixel Integration

- 10 x 10 sub-sampling grid
- Concentric sky annulus (4.0 - 7.0 px)
- MAD iterative sigma clipping:

$\sigma_{MAD} = 1.4826 \cdot \text{median}(|\Delta|)$

`1% Synthetic recovery verified`

3. CALIBRATION & SED

Mission v28 CBE Ground Truth

- Effective FWHM curve (4.61" - 6.87")
- Curve-of-growth flux correction
- 102-channel LVF array extraction
- Unit conversion: µJy, mJy, AB mag

`Full error propagation (Poisson+Sky)`

4. DISCOVERY & EXPORT

Multi-Epoch Differencing

- Multi-epoch blink comparator
- Signed difference dipole tagging
- Cross-match: Gaia DR3 & NEOSSat
- NASA FITS Standard 4.0 BINTABLE

`Validated by astropy & CFITSIO`

Application Architecture & Deployment Topology

CLIENT LAYER (React + TypeScript)

- Win32 Windows 2000 ergonomics (zero web bloat)
- HTML5 Canvas FITS visualizer with 8 colormaps
- Interactive reticle with real-time coordinate readout
- 1-4 epoch blinking animator (1-10 Hz)
- Spectral energy distribution modal via Chart.js

Deployment: Vercel + GitHub Pages

SERVER & PIPELINE (Node.js + C)

- Memory-bounded streaming FITS parser
- Concurrent worker pool for batch aperture jobs
- Headless CLI mode for automated shell scripts
- 2880-byte block padded NASA BINTABLE builder
- Automated disk cache indexing & local staging

Deployment: Hugging Face Spaces + Docker

DATA INTEROPERABILITY

- NASA/IPAC Infrared Science Archive (IRSA)
- ESA Gaia Data Release 3 (proper motions)
- CSA NEOSSat Near-Earth Asteroid orbits
- WISE / NEOWISE 15-year baseline baseline cross-links
- Formats: FITS BINTABLE, CSV, JSON, PNG

Open-source & 100% reproducible

SCIENTIFIC IMPACT & OUTREACH

Democratizing All-Sky Infrared Astrophysics

1. Citizen Science Discovery

- Brings professional-grade astronomical blink inspection to anyone with a browser.
- Zero installation or coding barrier: runs on Vercel, GitHub Pages, and Hugging Face.
- Inspect billions of sky positions across 4 survey epochs with 8 colormaps and reticle WCS.
- Enables crowdsourced candidate classification for faint asteroids and distant perturbers.

2. Undergraduate Educational Foundation

- Backed by a complete undergraduate research paper authored alongside the tool.
- Covers Keplerian celestial mechanics, thin-film Fabry-Perot optics, and blackbody physics.
- Provides clear mathematical derivations of difference imaging and Taylor-series dipoles.
- Serves as an open curriculum module for observational astrophysics students.

3. Cross-Survey Astronomical Synergies

- Extends a 15-year baseline by cross-linking SPHEREx multi-epoch data with NASA WISE (2010).
- At 400 AU, 15 years yields > 11 arcsec of motion (nearly 2 full SPHEREx pixels).
- Bridges optical surveys (Vera C. Rubin LSST) with continuous 102-channel infrared spectra.
- NASA-standard FITS BINTABLE outputs ensure immediate plug-and-play with Astropy pipelines.

OPEN SCIENCE & RESOURCES

Live Deployments, Source Code & Data Citations

Try the Live Platform & Inspect Source Code:

Web Application (Vercel):
https://spherex-photometry.vercel.app

Alternative Mirror (GitHub Pages):
https://yancovert.github.io/spherex-photometry/

Cloud Container (Hugging Face):
https://huggingface.co/spaces/Yancovert/spherex-photometry

Source Code (MIT License):
https://github.com/Yancovert/spherex-photometry

Research Paper:
spherex_research_paper.md / .tex / .docx (in repo)

Mandatory Data & Mission Citations:

- **NASA / IPAC Infrared Science Archive (IRSA):** SPHEREx Level 2 Calibrated Images, Public-products v28 Current Best Estimate (CBE) calibration tables, and TAP API.
- **ESA Gaia Mission (Gaia Collaboration):** Gaia Data Release 3 (DR3) astrometric proper motion catalogs and cross-match reference frames.
- **Canadian Space Agency (CSA):** NEOSSat (Near-Earth Object Surveillance Satellite) public orbital archives for asteroid verification.
- **Theoretical Framework:** Batygin & Brown (2016, 2021) 'Orbit of Planet Nine', AJ; Bock et al. (2018) 'SPHEREx All-Sky Spectral Survey'; Alard & Lupton (1998) 'Optimal Image Subtraction'.
- **Team Exulans:** Developed for NASA Space Apps Challenge 2026. Team logo designed by Freund0815.